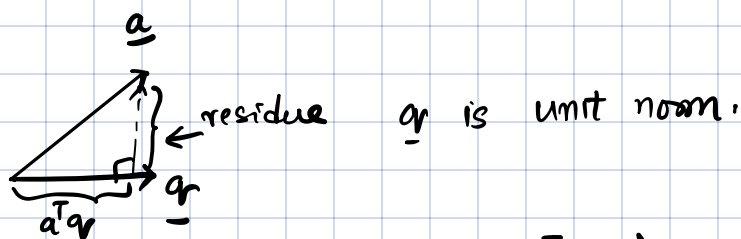


QR decomposition



Projection of $\underline{a}$ on $\underline{q}$ is $(\underline{a}^T \underline{q}) \underline{q}$

the error/residue vector $\underline{r} = \underline{a} - (\underline{a}^T \underline{q}) \underline{q}$ is perpendicular to $\underline{q}$

$$\underline{r}^T \underline{q} = \underline{a}^T \underline{q} - (\underline{a}^T \underline{q})(\underline{q}^T \underline{q}) = 0$$

$$\underline{a} = \underline{r} + \underline{q}$$

$$\langle \underline{a}, \underline{q} \rangle \longrightarrow \langle \underline{r}, \underline{q} \rangle$$

Given a set of K linearly independent vectors $\underline{a}_1, \underline{a}_2, \dots, \underline{a}_K$ in $\mathbb{R}^n$, How do we generate $\underline{q}_1, \underline{q}_2, \dots, \underline{q}_K$ that are orthonormal and $\text{Span}\{\underline{a}_1, \underline{a}_2, \dots, \underline{a}_K\} = \text{Span}\{\underline{q}_1, \underline{q}_2, \dots, \underline{q}_K\}$

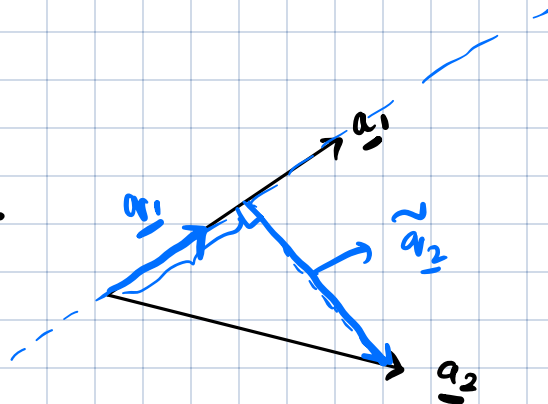
$$A = [\underline{a}_1 \ \underline{a}_2 \ \dots \ \underline{a}_K]$$

$$Q = [\underline{q}_1 \ \underline{q}_2 \ \dots \ \underline{q}_K]$$

$$\text{im}(A) = \text{im}(Q)$$

Algorithm outline

$$\underline{\tilde{q}}_1 = \underline{a}_1, \quad \underline{q}_1 = \frac{\underline{\tilde{q}}_1}{\|\underline{\tilde{q}}_1\|_2} = \frac{\underline{a}_1}{\|\underline{a}_1\|_2}$$



$$\underline{\tilde{q}}_2 = \underline{a}_2 - \underbrace{(\underline{a}_2^T \underline{q}_1) \underline{q}_1}_{\text{proj of } \underline{a}_2 \text{ along } \underline{q}_1}$$

$$\underline{q}_2 = \frac{\underline{\tilde{q}}_2}{\|\underline{\tilde{q}}_2\|_2}$$

$$\underline{a}_1 = \|\underline{a}_1\|_2 \underline{q}_1$$

$$\underline{a}_2 = (\underline{a}_2^T \underline{q}_1) \underline{q}_1$$

$$\|\underline{q}_2\|_2$$

$$+ \|\tilde{\underline{a}}_1\|_2 \underline{q}_2$$

$$\underline{q}_1 = \frac{\underline{a}_1}{\|\underline{a}_1\|_2}$$

$$\underline{q}_2 = \frac{\underline{a}_2}{\|\tilde{\underline{a}}_2\|_2} - (\underline{a}_2^T \underline{q}_1) \frac{\underline{a}_1}{\|\underline{a}_1\|_2}$$

Gram-Schmidt algorithm

Initialize $\tilde{\underline{q}}_1 = \underline{a}_1$, $\underline{q}_1 = \frac{\tilde{\underline{q}}_1}{\|\tilde{\underline{q}}_1\|_2}$ 3n operations

for $r = 2$ to K

$$\tilde{\underline{q}}_r = \underline{a}_r - \sum_{i=1}^{r-1} (\underline{a}_r^T \underline{q}_i) \underline{q}_i \quad \text{stage } r: \text{ currently processing } \underline{a}_r$$

already constructed $\underline{q}_1, \underline{q}_2, \dots, \underline{q}_{r-1}$

(r-1)(3n-1) + n(r-1)

If $\tilde{\underline{q}}_r = 0$, quit.

$$\underline{q}_r = \tilde{\underline{q}}_r / \|\tilde{\underline{q}}_r\|_2$$

output $\underline{q}_1, \underline{q}_2, \dots, \underline{q}_K$.

check:

① $\tilde{\underline{q}}_r$ is perpendicular to $\underline{q}_1, \underline{q}_2, \dots, \underline{q}_{r-1}$

Proof: By induction. Assume the statement above is true for $r-1$. To prove it is true for r .

Take inner product of both sides of ① with $\underline{q}_j$

for $1 \leq j \leq r-1$

$$\underline{q}_j^T \tilde{\underline{q}}_r = \underline{q}_j^T \underline{a}_r - \sum_{i=1}^{r-1} (\underline{a}_r^T \underline{q}_i) \underline{q}_i^T \underline{q}_j$$

only term $i=j$ survives.

$$\underline{a}_i^T \underline{a}_r = \underline{a}_i^T \underline{a}_r - (\underline{a}_i^T \underline{a}_j) (\underline{a}_j^T \underline{a}_r) \\ = 0$$

② Complexity of G-S

$$\begin{aligned} & \textcircled{3n} + \sum_{r=2}^k \left[(r-1)(3n-1) + n(r-1) \right] + \textcircled{3n} \\ &= 3n \sum_{r=1}^k 1 \approx 3Kn \quad (4n-1) \left[\sum_{r=2}^k (r-1) \right] \\ & \sim O(nk^2) \end{aligned}$$

③ $\underline{a}_i$ can be written in terms of $\underline{a}_1, \underline{a}_2, \dots, \underline{a}_i$

$\underline{q}_i$ can be written in terms of $\underline{a}_1, \underline{a}_2, \dots, \underline{a}_i$

$$A = \begin{bmatrix} \underline{a}_1 & \underline{a}_2 & \dots & \underline{a}_k \end{bmatrix}_{n \times k} = \begin{bmatrix} \underline{q}_1 & \underline{q}_2 & \dots & \underline{q}_k \end{bmatrix}_{n \times k} \begin{bmatrix} \boxed{1} & \boxed{0} & \dots & \boxed{0} \\ \boxed{0} & \boxed{1} & \dots & \boxed{0} \\ \vdots & \vdots & \ddots & \vdots \\ \boxed{0} & \boxed{0} & \dots & \boxed{1} \end{bmatrix}_{k \times k}$$

$\underbrace{\hspace{10em}}_{\text{all orthogonal}} \quad \underbrace{\hspace{10em}}_{\text{square upper triangular matrix}}$

$$\underline{k \leq n}$$

$$O(nk^2)$$

$$A = QR$$

④ G-S algorithm above executes to completion / never divides by 0.
Proof. Exercise.
 original k vectors are LI.

→ R is upper triangular

→ when the columns of A are LI, the diagonal entries in R

are +ve (non-zero).

Uses:

- ① Check linear independence.
- ② Solve linear equations.

$$A\underline{x} = \underline{b} \quad (\text{overdetermined})$$

$$\left\{ \begin{array}{l} \text{Compute } A = QR \quad O(nk^2) \\ \text{compute } \tilde{\underline{b}} = Q^T \underline{b} \quad O(nk) \\ \text{Solve } R\underline{x} = \tilde{\underline{b}} \text{ by } O(k^2) \\ \text{back-substitution} \end{array} \right. \quad O(nk^2)$$

$$\boxed{Q^T Q = I} \quad \checkmark$$

$$\left[\begin{array}{c} \\ \end{array} \right] \left[\begin{array}{c} \\ \end{array} \right]$$

(Round-off errors / precision errors)
(Error propagation)

$$\boxed{QQ^T \neq I}$$

$$\left[\begin{array}{c} \\ \end{array} \right] \left[\begin{array}{c} \\ \end{array} \right]$$

$$A = \left[\begin{array}{c|c} Q_1 & \\ \hline \end{array} \right]_{n \times n} \quad R_{n \times k}$$

$\underbrace{\hspace{10em}}_{n \times k \quad n \times (n-k)}$

Full-QR decomposition

$$A_{n \times k} = Q_{n \times k} R_{k \times k}$$

Thin-QR decomposition

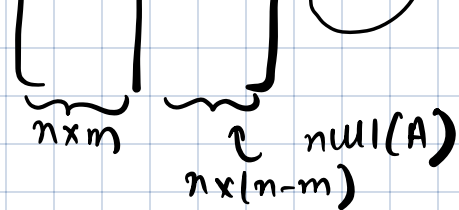
run full QR on A^T

$$A^T = \left[\begin{array}{c|c} Q_1 & Q_2 \end{array} \right] (R)$$

$$\left[\begin{array}{c|c|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \hline 0 & 0 & 1 \end{array} \right]$$

$m \times n$
fat

$n \times m$



$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{matrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{matrix}$$

rand()

$$a_4 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

QR decomposition

$$A = \underbrace{Q}_{n \times k} \underbrace{R}_{k \times k}$$

$k < n$

$$(Q^T Q = I)$$

$$\tilde{q}_4 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 0 \\ 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

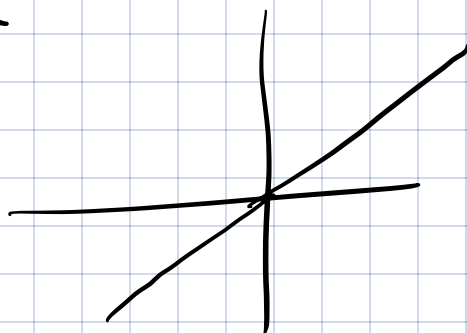
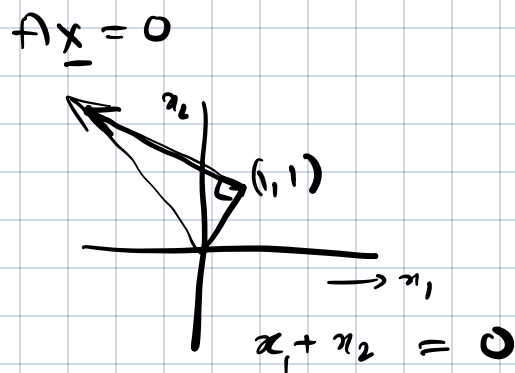
$$A_{n \times k} = \underbrace{Q}_{n \times k'} \underbrace{R}_{k' \times k}$$

Suppose A is a fat matrix. Can we find an orthogonal basis for $\text{null}(A)$?
 $A: m \times n$ ($m < n$)

Consider $A^T = Q R$

$n \times m$

$$\begin{bmatrix} \vdots \end{bmatrix} = \underbrace{\begin{bmatrix} Q \end{bmatrix}}_{n \times m} R$$



Thin QR: Find basis of image space of A

$$O(nk^2)$$

Full QR: Find basis of $\mathbb{R}^n$

$$\begin{bmatrix} q_1 \\ \vdots \end{bmatrix}$$

$$O(n^3)$$

$$x_1, x_2, \dots, x_n$$

Least-Squares
problem:

$$\min (x_1 + x_2 - 5)^2 + (2x_1 - 5x_2 + 10)^2 + (3x_1 - 20x_2 - 25)^2$$

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 2 & -5 \\ 3 & -20 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\underline{x}} \approx \underbrace{\begin{bmatrix} 5 \\ -10 \\ 25 \end{bmatrix}}_{\underline{b}}$$

When A is tall, $A\underline{x} = \underline{b}$ may not always have a solution.

Can we find an $\underline{x}$ that makes $A\underline{x}$ as close to $\underline{b}$ as possible.

$$\text{error} = A\underline{x} - \underline{b}$$

$$\text{Cost}(\underline{x}) = \|A\underline{x} - \underline{b}\|_2^2$$

$$A \text{ tall, } \min_{\underline{x}} \|A\underline{x} - \underline{b}\|_2^2$$

Suppose $A = V$ is tall orthogonal

$$\min_{\underline{x}} \|V\underline{x} - \underline{b}\|_2^2.$$

Solution:

$$\begin{aligned} \underbrace{\|V}_{3 \times 2} \underbrace{\underline{x}}_{2 \times 1} - \underbrace{\underline{b}}_{3 \times 1} \underbrace{\|_2^2}_{\text{norm}} &= \underbrace{(V\underline{x} - \underline{b})^T (V\underline{x} - \underline{b})}_{\text{dot product}} \\ &= \underbrace{\underline{x}^T V^T V \underline{x}}_{\text{identity}} - \underbrace{\underline{b}^T V \underline{x}}_{\text{dot product}} - \underbrace{\underline{x}^T V^T \underline{b}}_{\text{dot product}} + \underbrace{\underline{b}^T \underline{b}}_{\text{dot product}} \\ &= \underline{x}^T \underline{x} + \underline{b}^T \underline{b} - 2 \underline{b}^T V \underline{x} \end{aligned}$$

$$\{x_1^2 + x_2^2 + b_1^2 + b_2^2 + b_3^2 - 2x_1 y_1 - 2x_2 y_2\}$$

$$x_1 = y_1$$

$$\begin{pmatrix} y_1 \\ b_1 \end{pmatrix} = \underbrace{V^T b}_{2 \times 1}$$

$$\rightarrow \underbrace{(x_1 - y_1)^2}_{\uparrow} + \underbrace{(x_2 - y_2)^2}_{\uparrow} + \underbrace{b_1^2 + b_2^2 + b_3^2 - y_1^2 - y_2^2}_{\uparrow}$$

$$\text{Assume } V \text{ is tall orthogonal} \quad \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \underbrace{V^T b}$$

$$Vx = b$$

$$\textcircled{1} \quad \underline{x} = V^T \underline{b} \text{ is the solution to } \min_{\underline{x}} \|V\underline{x} - \underline{b}\|^2.$$

$$\underline{x} = V^T \underline{b}$$

$$\textcircled{2} \quad V V^T \underline{b} \text{ is the projection of } \underline{b} \text{ on } \text{im}(V). \\ (I - V V^T) \underline{b}$$

$$\text{In general for a matrix } A = \underbrace{Q}_{n \times k} \underbrace{R}_{k \times k} \quad (\text{columns of } A \text{ LI}).$$

$$\textcircled{1} \quad \text{solution to } \min_{\underline{x}} \|A\underline{x} - \underline{b}\|$$

$$A = QR \\ A^T = R^T Q^T$$

$$\equiv \min_{\underline{y}} \|Q\underline{y} - \underline{b}\|^2 \\ \underline{y} = Q^T \underline{b}$$

$$\textcircled{2} \quad \text{Vector in } \text{im}(A) = \text{im}(Q) \text{ that is closest to } \underline{b} \\ \text{is } Q Q^T \underline{b}$$

$$\text{To find } \underline{x}, \text{ we do: } A\underline{x} = Q\underline{y} \\ QR\underline{x} = QQ^T \underline{b}$$

$$R\bar{x} = Q^T b$$

To solve $\min_{\bar{x}} \|A\bar{x} - \underline{b}\|^2$, we do the following

(i) compute $A = QR$ (thin-QR)

(ii) compute $Q^T b$

(iii) solve $R\bar{x} = Q^T b$ by back-substitution

Identical to the steps used to solve $A\bar{x} = \underline{b}$.

the least squares solution obtained above can also be obtained by solving $A^T A \bar{x} = A^T \underline{b}$.

(d) A has linearly independent columns;



$A^T A$ is full rank.

(a) $A\bar{x} = 0 \iff A^T A \bar{x} = 0$

(b) $\text{null}(A) = \text{null}(A^T A)$

(c) $\text{rank}(A) = \text{rank}(A^T A)$

(e) if A has linearly indep columns,

$A\bar{x} = \underline{b}$

$A^T A \bar{x} = A^T \underline{b}$

(f)

Pseudo inverse of A .

$$\underline{x} = \underbrace{(A^T A)^{-1} A^T}_{\uparrow} \underline{b}$$
$$\underbrace{(A^T A)^{-1} A^T}_{\uparrow} A = I.$$

$$A = QR$$

$$A^T = R^T Q^T$$

$$(A^T A)^{-1} = (R^T R)^{-1} = \bar{R}^{-1} \bar{R}^T$$

$$(A^T A)^{-1} A^T b = \bar{R}^{-1} \cancel{\bar{R}^T} \cancel{Q^T} b$$

$$= \bar{R}^{-1} Q^T b$$

Read:

Section I, II from

VMIS

(Stephen Boyd)

Same Solution obtained
by least squares
above.